

DEXTER FERNANDES

Senior Data Scientist Bristol, UK +44 7436 007369 dexterfernandes11@gmail.com linkedin.com/in/dexterferns
github.com/Dexter-Fernandes

SUMMARY

Senior data scientist and computer vision engineer with 3+ years building production machine learning systems for industrial inspection, anomaly detection and operational analytics. Owns multimodal data pipelines, statistical experimentation, MLOps and deployment across Microsoft Azure and edge platforms. Led four engineers and delivered real-time perception using PyTorch, TensorFlow, scikit-learn and NVIDIA Jetson. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar - AI Engineer

01/2023 - 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, adapting YOLOv5 in PyTorch and taking models from data curation through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Built ingestion and validation pipelines across image, video and sensor data with Airflow, MLflow, DVC and CVAT, processing 3.6 million records and cutting rollout from 72 hours to under 24 hours.
- Designed adversarial-condition experiments for occlusion, lighting and motion artefacts, using Optuna, Weights & Biases, Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals to guide model selection.
- Owned production monitoring and drift tracking for fielded vision systems, using TIDE error analysis on COCO-format data to direct threshold tuning, dataset improvements and model retraining.
- Hardened DeepStream edge services with Dockerised microservices, Kafka, structured logs, health checks, timeouts and safe restarts, cutting live video pipeline crashes by 67% and enabling remote recovery.
- Built activity classification over 91 sensor channels using MultiRocket, gradient boosting and random forests for multivariate time-series analysis.

LivNSense Technologies - Lead Software Engineer, AI

05/2020 - 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector and helmet-colour classifier for construction safety in Texas, reducing false positives by 33% within edge latency and power budgets while setting model-evaluation standards.
- Owned production ML pipelines from ingestion and labelling through training, optimisation, deployment, monitoring and field validation, coordinating hardware, software and QA teams for industrial safety rollouts.
- Built YOLOv5 visual quality inspection for John Deere engine paintwork, detecting defects, contamination and colour inconsistency, and evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for segmentation-based review.
- Automated labelling with existing detection models and DVC dataset versioning, cutting manual labelling effort by 80%; synthetic-data tooling reduced further manual work by 60%.
- Deployed containerised inference microservices on Azure with REST endpoints and queue-based processing, owning autoscaling, monitoring and cost-per-inference while evaluating edge-cloud bandwidth and latency trade-offs.
- Optimised C++17 OpenVINO pipelines with asynchronous inference, FP16/INT8, quantisation-aware training and structured pruning, raising throughput from 5 to 15 FPS and halving model size while holding production accuracy.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built a DeepStream 9.0 and GStreamer pipeline for YOLO detection and NvDCF tracking across three concurrent RTSP feeds, with metadata extraction, anonymisation and re-streaming.
- Converted YOLO26n to a TensorRT FP16 engine supporting multiple batch sizes, validating accuracy against FP32 and profiling p99 latency to sustain 29.7 FPS per stream on 6GB VRAM.
- Implemented a C++ and CUDA TensorRT decode plugin and compared IoU, NvDCF and NvSORT trackers on MOTA, IDF1, identity switches, fragmentation, throughput and VRAM impact to guide selection under edge constraints.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a ROS2 Jazzy and Gazebo Harmonic robotics stack to evaluate LiDAR SLAM, LiDAR and RGB fusion, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations to compare mapping and localisation behaviour.
- Developed C++ and Python nodes for point-cloud filtering, registration and EKF localisation using PCL, Open3D and robot_localization, debugging transform timing through tf2 and timestamp alignment. Containerised the stack with Docker for reproducible Linux builds.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR, then extended the harness to EfficientViT, MobileSAM and Depth Anything v2 to assess detection, segmentation, depth and open-vocabulary approaches under the same accuracy, latency and model-size criteria.
- Exported models to ONNX and evaluated OpenVINO deployment paths, comparing mAP, inference time and model size through a repeatable harness. Kept preprocessing, measurement and reporting consistent across model families to support engineering decisions about accuracy and latency trade-offs.

EDUCATION

University of Bristol - MSc Robotics, Merit

09/2023 - 09/2024 · BRISTOL, UK

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time inspection-mapping system in C++17 and ROS2 on a Clearpath Husky, combining VLP-16 LiDAR, RealSense RGB-D and radiological sensing.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer under a common protocol, configured Ceres, g2o and GTSAM solver backends, and calibrated LiDAR extrinsics through tf2 and PCL pipelines.
- Ran the perception stack on an NVIDIA Jetson AGX Xavier and configured Nav2 for autonomous waypoint surveys.

Visvesvaraya National Institute of Technology - BTech Electrical and Electronics Engineering

07/2016 - 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control for real-time AI, sensing and robotics.

SKILLS

Languages	Python, C++17, CUDA C++, SQL, Bash
ML and statistics	PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Bayesian optimisation, hypothesis testing, confidence intervals, ablation studies
Computer vision	Object detection, image classification, semantic and instance segmentation, anomaly and defect detection, object tracking, optical flow, OpenCV
Data and MLOps	MLflow, DVC, Apache Airflow, Apache Kafka, CVAT, model monitoring, drift tracking, retraining, automated labelling, experiment tracking
Edge AI	NVIDIA DeepStream, GStreamer, TensorRT, OpenVINO, ONNX, FP16/INT8, NVIDIA Jetson AGX Xavier, quantisation, pruning, latency and throughput profiling
Cloud and production	Microsoft Azure, Docker, CI/CD, REST APIs, containerised microservices, PostgreSQL, Django, observability, Linux
Robotics and sensing	ROS2 (Robot Operating System 2), LiDAR SLAM, Nav2, RTAB-Map, PCL, Open3D, tf2, sensor fusion, autonomous navigation
Leadership	Technical leadership, mentoring, model validation, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage - Technical Assessor, Computer Vision and Software Engineering

05/2026 - PRESENT · REMOTE

Assesses candidate depth across computer vision, perception, SLAM and deployment.

Taco Bell - Service Champion / Team Trainer

09/2024 - PRESENT · BRISTOL, UK

Trains new staff and handles customer escalations in a high-volume service environment.

CERTIFICATION

Deep Learning Specialisation, deeplearning.ai (07/2019)